ZOO506-Applied-Entomology-And-Pest-Management

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Explain and Name the types of four stages insect larvae.

About 75% of all insect species go through the four stages of complete metamorphosis - egg, larva, pupa, and adult. The larva is a specialized feeding stage that looks very different from the adult.

1) The Egg: Majority of insects begin their life in fertilized eggs. Egg laying amongst insects depends on the species it belongs to, for example, cockroaches or mantids lay their eggs in a bunch attached together that forms the ootheca or egg packet.

2) The Larvae: Insect larvae have several distinct ways of hatching. While caterpillar larvae bite their way out of the eggs during the hatching process. The insect larva undergoes several stages of molting where it sheds old cuticles multiple times to get new cuticles. Insect larvae vary in shape depending on the species, therefore they are divided into five kinds. The caterpillar-like larvae are eruciform, the maggot-like larvae are vermiform, the wire worm-like larvae are elateriform, the grub-like larvae are scarabaeiform and the long, elongated larvae are campodeiform. These larvae then transform into pupa and enter the next stage.

3) The Pupa: During this stage, the pupa is mostly resting and inactive. The larval tissues collapse and reorganize the adult form of the pupa. After this process is over, the pupa molts and emerges out as an adult with a pair of wings. In the pupa stage, the wings develop into its complete form and the adult insect enters the world.

4) The Adult: Adult insects have tactile hair, which is found on the antennae, legs, torsal regions of the body, and helps in signaling the insect to its external environment. Tactile hair covers the entire surface of an insect. In this stage, the adult insect is sexually mature and spends its time mostly reproducing. The mating in adult insects vary depending upon the species.

Name the parts of thorax and abdomen of insect.

Thorax:

The thorax, comprises three large segments:

- Prothorax (anterior most segment),
- Mesothorax (middle), and
- Metathorax (posterior).

Abdomen:

The basic number of segments in the abdomen is eleven plus the telson which bears the anus. Each segment of the abdomen consists of a

- Dorsal sclerite, the tergum, and a
- Ventral sclerite, the sternum,

These joined to one another laterally by a pleural membrane. On the abdomen are various organs such as the ovipositor, sting, cerci and styli.

How allergy is caused?

Allergy (hay fever, asthma, etc.) are frequently caused by exposure of susceptible persons to dried insect scales and hairs.

Define disease with an example:

Any harmful deviation from the normal structural or functional state of an organism is called Disease.

Example: the Yellow Fever and the Malaria is the best examples of the disease.

What are tracheal gills?

Insects which in their early stage of live in water cannot, of course, breathe air into their bodies through spiracles during that period of their live. There are enclosed in such cases and the animal obtains and usually through special structures called tracheal gills.

What are the Different characteristics to identify the insect./ characteristics of insects.

1= Bilaterally symmetrical. 2= Consisting of a series of segments. (a) Head (b) thorax (c) abdomen. 3= The animal is covered by a Shell like skeleton present outside the body, horny in nature. 4= Three pairs of jointed legs attached to the thorax. 5= Pair of antenna. 6= Mouth parts. 7= Two pair of wings. 8= Breathing through Air tubes. 9= Reproductive organs present near the hinder end of the body.

What are the Insect-borne diseases:

Insect-borne diseases are viral and bacterial illnesses from insect (bug) bites.

- mosquitoes are known for spreading the Zika virus, Yellow Fever, and Malaria.
- Malaria (protozoan): Anopheles species of mosquito.
- Lymphatic filariasis (nematode worm): Culex, Anopheles, Aedes species of mosquito.
- Dengue (virus): Aedes species of mosquito.
- Leishmaniasis (protozoan): mainly Phlebotomus species of sandfly.

Types of insect head:

Based on the inclination of long axis of the head and orientation of mouth parts there are three types of insect heads.

- Hypognathous (Hypo – Below: Gnathous – Jaw)
- Prognathous : (Pro – infront: Gnathous – Jaw)
- Opisthognathous : (Opistho – behind: Gnathous – Jaw)

What are the Four types of larvae:

1= Larvae of butterflies and moths are commonly called **caterpillars**, 2= Those of beetles are usually called **grubs**, 3= Those of flies are called **maggots**, 4= Larvae found boring in wood, however, whether they will become moths, beetles or other insects, are uniformly called **borers.**

Describe nematode disease and mention its vector name and its pathogen?

A number of genera and species of nematodes are highly damaging to a great range of hosts, including foliage plants, agronomic and vegetable crops, fruit and nut trees, turfgrass, and forest trees. Some of the most damaging nematodes are: Root knot (Meloidogyne spp.); Cyst (Heterodera and Globodera spp.); Root lesion (Pratylenchus spp.); Spiral (Helicotylenchus spp.); Burrowing (Radopholus similis); Bulb and stem (Ditylenchus dipsaci); Reniform (Rotylenchulus reniformis); Dagger (Xiphinema spp.); Bud and leaf (Aphelenchoides spp.); and Pine Wilt Disease (Bursaphelenchus xylophilus). The only known nematode vectors are in the genera Xiphinema and Longidorus.

Differentiate between pulivillus, pulvilli, and epodium ?

Pulivillus: a soft, padlike structure located at the base of each claw on the feet of certain insects.

Pulvilli: Pulvilli are lobes or padds that are located between the tarsal claws of many insects.

Empodium: Between the pulvilli there is often a spine that is called the empodium.

Physical transmission:

Physical transmission or the mechanical transmission, the simple transfer of agents from one infected host or a contaminated substrate to a susceptible host occurs.

Nervous system of insects:

The nervous system of insects is located along the middle line of the body quite near its under surface As in animals generally, it is composed of cells and fibers Ganglia: former are for the most part gathered together in clusters which are called ganglia And from each of the cells in a ganglion one or more nerve fibers pass out, to connect either with some other nerve cell or with some structure of the body The larger nerves are really bundles of the fibers running side by side like the wires of a telephone cable Apparently each segment of the insect body once had but with the fusion of the segments tube of a fly Greatly many of these have also fused, reducing the separate ganglia in adult insects to a smaller number, which varies central in different kinds This fusion has been produced by the hinder ganglia moving forward until in some cases none is found in the abdomen Each ganglion is connected to the one in front and the one behind by one or two bundles of nerve fibers which are called commissures Each consists of numerous fibers which taken together form the means of communication between the different parts of the system.

What are the parts of insect leg:

Legs of insects (adults and larvae) are jointed and have five parts: 1= coxa, 2= trochanter, 3= femur, 4= tibia, and 5= tarsus.

Name head positions in insect:

Heads with the mouth beneath are called hypognathous those with it in front are prognathous. Structures found on the head are

- A pair of antennae
- Two compound eyes (ocelli)
- The mouth parts

Difference between mollusks and arthropods exoskeleton:

The arthropods have exoskeleton covers on their body. This is generally made of chitin. The Arthropoda is the largest phylum of the kingdom Animalia. Example: Spiders, Insects.

Molluscs have their soft bodies covered and protected by hard shells. The Mollusca is the second largest or diverse phylum in the animal kingdom. Example: Snails

Define ovipositor:

Egg-laying tubular structure at the end of the abdomen in many female insects and some fishes.

Define coxa:

The coxa is the first (counting from the body) segment in the leg of an insect. The coxa attaches the leg to the thorax.

What is meant by contact insecticides?

It was formerly supposed that certain materials called contact insecticides, which kill insects by contact with their bodies. It is caused death by entering the spiracles and closing them up, thus producing suffocation.

What is cerci?

A pair of antenna like structure (sometimes short sometimes long) present at the end of abdomen are known as cerci.

Explain mouthparts with regard to possition:

- **Labrum** - a cover which may be loosely referred to as the upper lip.
- **Mandibles** - hard, powerful cutting jaws.
- **Maxillae** - 'pincers' which are less powerful than the mandibles.
- **Labium** - the lower cover, often referred to as the lower lip.
- **Hypopharynx** - a tongue-like structure in the floor of the mouth.